Phu Nguyen

Going into this lab, I honestly thought data prep was just a quick cleanup step before the "real" machine learning started. But after going through it, I see it's more like getting ingredients ready before you cook, because if the data isn't ready the model can't even use it. The part that helped me most was filling in Age with the median. I used the median instead of the mean because a few really old passengers can pull the average up, and the median doesn't get messed up by that. The one-hot encoding was also interesting, since models only understand numbers and not words like male. Fare goes up to 512 and Age only goes to about 80, so without scaling the model might think Fare matters more just because the numbers are bigger. I also learned that you only scale the number columns, not the encoded ones. And a decision tree doesn't need scaling at all because it only looks at the order of the values in one column at a time. Overall this lab made data prep feel a lot more important than I expected, and I can see why it's such a big part of the ML workflow.

Cyndy Ndong

Working through Lab 05 helped me see that data preprocessing is the foundation of machine learning, not just administrative setup. Handling the Titanic dataset step-by-step showed me how much cleaner data impacts a model's ability to learn.

A major key takeaway was learning why we use the median instead of the mean for missing values in columns like Age. The median keeps the central value true even if extreme ages exist, preventing artificial skews. I also learned why machine learning models need One-Hot Encoding for categorical features like Sex and Embarked. Algorithms are mathematical, so converting text into binary ones and zeros allows the model to process categories without assuming a false ranking between them.

Additionally, applying StandardScaler to Age and Fare made it clear why feature scaling matters.

Scaling puts all numerical features on equal footing so larger values do not dominate distance calculations. My biggest technical challenge was fixing syntax errors from squished code lines in Jupyter Lab, which taught me to review my Python code carefully line-by-line. Overall, this lab built my confidence in using Pandas and Scikit-Learn to prepare raw data for machine learning models.